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Gradient Boosting

Exp	Degree	Salary (y)	$\hat{y}$	Corrected values R_2	(actual - pred) R_1
2	BE	50k	75k	-23	-25
3	MASTERS	70K	75k	-3	-5
5	MASTERS	80k	75k	3	5
6	PHD	100k.	75k	20	25

Step 1) Base model → 1
o/p

↳ avg of Salary
$\Rightarrow \frac{50+70+80+100}{4} \Rightarrow$ 75K.

Step 2) Compute Residuals, Errors, Pseudo Residuals.

Step 3) Construct Decision Tree.
$\{x_i, R_1\}$

$$F(x) = h_0(x) + \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_n h_n(x)$$

$h \to$ o/p

$$= \left\{ \sum_{i=1}^{n} \alpha_i h_i(x) \right\}$$

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Math intuition behind Gradient boosting.

Things required to use GB

① I/P variables — dependent
$\{x_i, y_i\}$
independent

② $L(y, f(x))$

③ No. of Tree.

① Initialize model with Const value.

$$F_0(x) = \arg\min_{\gamma} \left(\sum_{i=1}^{n} L(y, \gamma)\right)$$

gamma
↳ Predicted O/P.

Iterate

② $m = 1$ to M

① Compute Pseudo Residuals.

$$\left\{ r_{im} = -\left[\frac{\partial L(y, F(x_i))}{\partial F(x_i)}\right] \right\}$$

② Fit a Base Learner $h_m(x)$
i/p $\{(x_i, r_{im})\}$

③ $$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, F_{m-1}(x_i) + \gamma)$$

④ Update Model

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

$$\left\{ Loss = \sum_{i=1}^{n} \frac{1}{2}(y_i - \hat{y})^2 \right\}$$

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Step ①

Exp	Degree	Salary	$\hat{y}$
2	BE	50K	60
3	PHD	70K	60
4	MASTERS	60K	60

$$= \frac{1}{2}(50-\hat{y})^2 + \frac{1}{2}(70-\hat{y})^2 + \frac{1}{2}(60-\hat{y})^2$$

Calculating 1st order derivative wrto $\hat{y}$

$$\frac{2}{2}(50-\hat{y})\cdot(-1) + \frac{2}{2}(70-\hat{y})(-1) + \frac{2}{2}(60-\hat{y})(-$$

$$\Rightarrow -180 + 3\hat{y} = 0$$
$$3\hat{y} = 180$$
$$\hat{y} = 180/3$$
$$\boxed{\Rightarrow 60}$$

Base Model.

Step ②

$$r_{im} = y - \hat{y} \;\div\; \left[-y + \hat{y} \text{ (1st order der. of loss func)} \right]$$

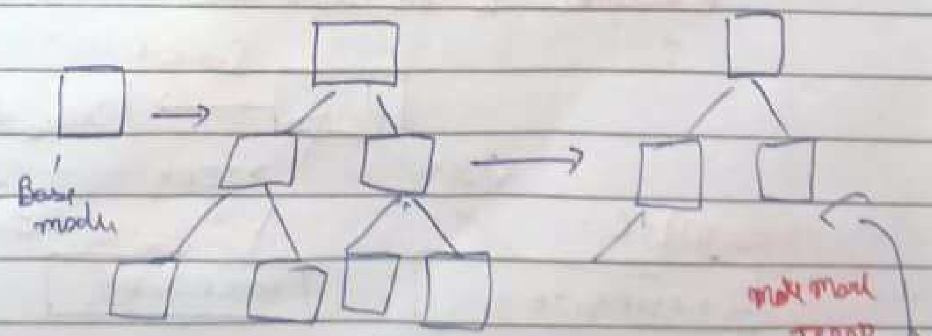
Residual values (error calc^n)

$$\Rightarrow r_{11} = (50-60) = \boxed{-10}$$
$$r_{21} = (70-60) = \boxed{10}$$
$$r_{31} = (60-60) = \boxed{0}$$

Step ③ γ_m

$$\gamma_1 = \sum_{i=1}^{n} \frac{1}{2}((y_i) - (60 + \hat{y}))^2.$$

Gen. value of loss !!



$\alpha = 0.1$

60 + (−10) y=1

60 + α (−10)

close but

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XG BOOST in classification

or

EXtreme Gradient boost!!

calculate after Post Pruning

f_1 feature¹ Salary	f_2 feature 2 Credit	f_3 feature 3 Approval	O/P. Rs (A_σ-P_r)	New P_r.	Rs
<=50k	B (Bad)	0	-0.5	0.6	-0.6
<=50k	G (Good)	1	0.5	0.4	0.6
<=50k	G	1	0.5	0.3	0.7
~~>~~=50k	B	0	-0.5	0.7	-0.7
~~>~~=50k	G	1	0.5	0.2	'
~~>~~=50k	N (Normal)	1	0.5	0.1	'
<=50k	N	0	-0.5		

Steps

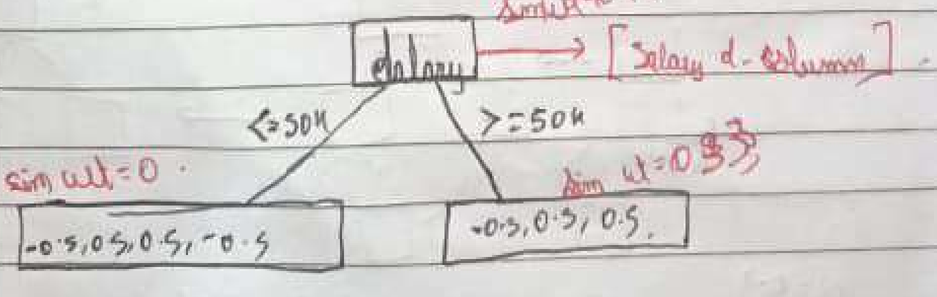
① Construct Base model.

∵ Classif. model, Only 2 O/P → 0 or 1

$\Rightarrow \frac{0+1}{2} \Rightarrow \frac{1}{2} = 0.5$

$\{P_r = 0.5\} \rightarrow$ Base.

① Construct a Tree with root.



② Calculate Similarity Weight

$$\Rightarrow \left\{ \frac{\sum (\text{Residual})^2}{\sum (P_r(1-\text{Prob}) + \lambda} \right\}$$

ⓐ · Sal <= 50k.

$$\Rightarrow \frac{[-0.5+0.5+0.5-0.5]^2}{[0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)]} = \frac{0}{0.25 \times 4 \Rightarrow 1} = 0$$

∴ Sal >= 50

$$\frac{[-0.5+0.5+0.5]^2}{3(0.5(1-0.5))} \Rightarrow \frac{0.25}{0.75} \quad \frac{1}{3} = 0.\overline{33}$$

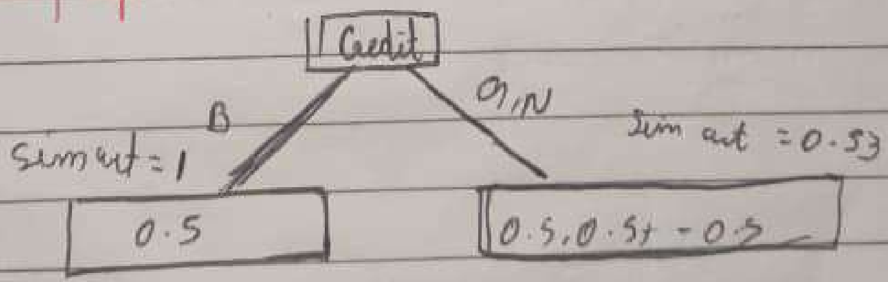
Gain = Σ sim. wt(node) − sim wt(root)

$\Rightarrow 0 + 0.33 - 0.14$

$\Rightarrow$ ~~0.478~~

· 0.19

⇒ making split wrt Credit (BT).



$$\left\{ \begin{aligned} \text{Gain} &= 1 + 0.33 - 0 \\ &\quad 1.33 \end{aligned} \right\}$$

Post Pruning

↳ This is done to figure if we need to split the tree further

$$\sigma(0 + \underbrace{\alpha}_{\text{0 to 1 (assume 0.1)}}(1))$$

sim - wt of B in split.

↳ 0 to 1 (assume 0.1)

$\sigma(0 + 0.1(1))$

$\sigma(0.1)$

$\Rightarrow \dfrac{1}{1 + e^{-0.1}} \Rightarrow \dfrac{1}{1 + \frac{1}{e^{0.1}}} \Rightarrow \dfrac{e^{0.1}}{e^{0.1} + 1}$

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XG BOOST Regressor.

Avg sal = 51K, Base model = 51K.

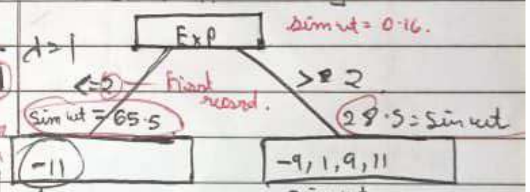
Exp	Gap	Salary ($)	Residual 1
2	Yes	40K	-11K (40-51)
2.5	Yes	42K	-9K
3	No	52K	1K
4	No	60K	9K
4.5	Yes	62K	11K

Similarity weight = $\frac{\sum (\text{Residual})^2}{\sum P_r(1-P_r)}$ → Used to classify how we have to split the tree!!
for classif.

Similarity weight for Regressor = $\frac{\sum (\text{Residual})^2}{\text{Num. of Res} + \lambda}$ λ → Hyper parameter

$\lambda = 1$

[-11, -9, 1, 9, 11]

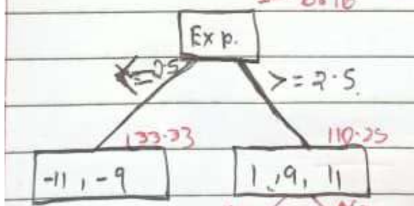


sim for this !!
$\lambda = 1$
⇒ $\frac{121}{1+1}$ ⇒ $\frac{121}{2}$ ⇒ 65.5

↳ $\frac{144}{4+1}$ ⇒ 28.5

Gain = 65.5 + 28.5 - 0.16
⇒ 93.84.

si = 0.16



sim = $\frac{(-11-9)^2}{2+1}$ ⇒ $\frac{400}{3}$ ⇒ 133.33.

O/P 11

sim = $\frac{(21)^2}{3+1}$ O/P = 5

⇒ 110.25

Gain = 133.33 + 110.25 - 0.16
⇒ 143.42

This split would be preferred on 2.5

Sim do for